

Research Article

FPGA-Based Real-Time Embedded Fish Embryo Detection System

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In today's aquaculture industry, the period of fish embryos needs to be detected in real time because different periods of fish embryos require different environments for their cultivation. The paper proposes an FPGA-based real-time embedded fish embryo detection system to solve the problems of today's fish embryo period detection, which consumes a lot of human resources and has low accuracy. Based on the selection of YOLOv3 as the basic model, this paper combines model optimization and hardware acceleration to make the whole system achieve the effect of real-time detection. With only a 4.23% reduction in the mAP value of the model, the inference speed of the model was improved by nearly 150%. Finally, we achieved 34.1 fps of real-time fish embryo detection on a 100 MHz FPGA chip, while having 82.49% mAP, and the whole embedded system only consumed 2.524w while running. The results show that the FPGA-based real-time fish embryo detection system proposed in this paper is fully capable of meeting the needs of today's aquaculture embedded devices. It is beneficial to promote the transformation of fish embryo analysis from traditional manual methods to AI-based embedded devices and facilitate the rapid analysis of fish embryos.

1. Introduction

Fish embryonic development has always been an extremely critical aspect of aquaculture. It is crucial for the continuation of rare fish species, high yield breeding, and sustainable development of fishery resources [1]. Since a mismatched culture environment can have a significant impact on embryo development, mainly in terms of survival rate and malformation rate, real-time fish embryo detection results are needed to adjust the corresponding culture environment [2–4]. Because fish embryos are very small before they are fully developed, the traditional method is to manually operate a microscope to observe the growth of fish embryos [1]. The approach has many problems, such as the difficulty of deploying equipment in fish culture greenhouses, the experience gap between analysts that affects the accuracy of identification, and the difficulty of analyzing directly in the culture environment. Therefore, automation and miniaturization of equipment are the main research directions for fish embryo detection in the future [5, 6].

It is difficult to achieve good results in aquaculture by relying only on embryo analysis by human eyes, so it is clearly not feasible to spend a lot of human resources to perform embryo analysis. Artificial intelligence would be a good solution to solve the problem in traditional methods. In 2017, Ishaq et al. demonstrate the potential of a deep learning approach for accurate high-throughput classification of whole-body zebrafish [7]. In 2019, Rauf et al. develop a framework based on improved CNN to classify the fish species [8]. In 2021, Naderi et al. develop CNN-based framework for cardiac function assessment in embryonic zebrafish from heart beating videos [9]. With the widespread use of artificial intelligence in cytology, bioinformatics, and medicine [10–12], embryo detection in aquaculture and academia is gradually experimenting with the approach. Qian et al. [13] established a microinjection system for detecting zebrafish embryos using CNN. However, the method is more suitable for deployment in the laboratory and cannot be directly applied to aquaculture. Xu et al. [14] built a Drosophila embryo detection framework using multifeature fusion (MFF) CNN on GPU. Dirvanauskas